

# Harmonising Kanta lab codes to OMOP

From a local **TEST\_NAME** + **UNIT** to a LOINC-backed OMOP concept

A six-step pipeline, an LLM that has to say *why*,  
and an honest score against a hand-curated reference

# Input data: v3 → v4

Every TEST\_NAME / UNIT pair against the three things it may carry:  
a real name, a recorded unit, and at least one record with a value.

## v3

| name  | unit | value | n rows | % rows | n records   | % rec        |
|-------|------|-------|--------|--------|-------------|--------------|
| X     | X    | X     | 8,606  | 32.3%  | 179,527,254 | <b>69.8%</b> |
| X     | X    |       | 150    | 0.6%   | 17,034      | 0.0%         |
| X     |      | X     | 3,541  | 13.3%  | 61,271,567  | 23.8%        |
| X     |      |       | 14,316 | 53.8%  | 16,547,150  | 6.4%         |
| total |      |       | 26,613 | 100%   | 257,363,005 | 100%         |

## v4

| name  | unit | value | n rows | % rows | n records   | % rec        |
|-------|------|-------|--------|--------|-------------|--------------|
| X     | X    | X     | 8,745  | 32.6%  | 212,508,947 | <b>82.4%</b> |
| X     | X    |       | 170    | 0.6%   | 31,102      | 0.0%         |
| X     |      | X     | 2,928  | 10.9%  | 10,479,927  | 4.1%         |
| X     |      |       | 14,955 | 55.8%  | 34,795,820  | 13.5%        |
| total |      |       | 26,798 | 100%   | 257,815,796 | 100%         |

DATA\_v3/1\_GetSummaryData/labSummaryStats.md · DATA\_v4/1\_GetSummaryData/labSummaryStats.md

## Input data: v3 → v4

**More records now carry a unit.** The share of records with name + unit + value rises from **69.8% → 82.4%**, and the "value but no unit" bucket collapses from 23.8% to 4.1% of records.

The row counts barely move (26,613 → 26,798). What changed is not *which codes exist* but *how much each code tells us about itself* — and the unit is the single field that decides a measurement's quantity.

Still open: **55.8% of rows carry neither a unit nor a value** (13.5% of records). Those rows are the hard part of everything that follows.

# Step 1 — The initial data

Five codes that will follow us through the whole deck.

| TEST_NAME | UNIT     | n         | value_missing_p | value_deciles                                        |
|-----------|----------|-----------|-----------------|------------------------------------------------------|
| p-na      | mmol/l   | 7,320,578 | 0.03            | 134.0, 136.4, 138.0, 139, 139.9, 140, 141, 142, 143  |
| p-na      | (none)   | 81,059    | 100             | (none)                                               |
| u-gluk-o  | (none)   | 658,060   | 100             | (none)                                               |
| ab-ca-ion | mmol/l   | 52,238    | 0               | 1.00, 1.04, 1.07, 1.10, 1.12, 1.14, 1.16, 1.18, 1.22 |
| b-hba1c   | mmol/mol | 1,976,825 | 0               | 33.1, 35.1, 36.9, 38.4, 40.0, 42.0, 45.1, 50.3, 60.3 |

Note rows 1 and 2: the same **TEST\_NAME** , split by unit. The pipeline maps (TEST\_NAME, UNIT) pairs, not names — because p-na in mmol/l and p-na with no unit are not the same measurement as far as LOINC is concerned.

Source: DATA\_v4/1\_GetSummaryData/labSummary.tsv · 26,798 rows

## Step 2 — Append what we already know

Before asking a model anything, attach every fact that already exists: the official long name from **Koodistopalvelu**, and the meaning of the **prefix / suffix** from the Finnish lab-code convention document.

| TEST_NAME | UNIT     | LongName            | prefix_meaning | suffix_meaning                            |
|-----------|----------|---------------------|----------------|-------------------------------------------|
| p-na      | mmol/l   | P -Natrium          | Plasma         | Native preparation                        |
| p-na      | (none)   | P -Natrium          | Plasma         | Native preparation                        |
| u-gluk-o  | (none)   | U -Glukoosi (kval)  | Urine          | Qualitative test (also semi-quantitative) |
| ab-ca-ion | mmol/l   | (none)              | Arterial blood | Ionized                                   |
| b-hba1c   | mmol/mol | B -Hemoglobiini-A1c | Blood          | (none)                                    |

This is where most of the signal comes from. `u-gluk-o` has **no unit and no values** — but `(kval)` in the long name and `-o` in the suffix dictionary say *qualitative*, which is enough to fix the whole measurement.

Source: `DATA_v4/2_AppendKnownInformation/knownInformation.tsv`

# Step 3 — Group the codes by string similarity

Codes are clustered on string distance so that near-identical spellings of the same test land in one prompt, and the model sees them side by side.

| step                                                | value                              |
|-----------------------------------------------------|------------------------------------|
| Rows in                                             | 26,798                             |
| Minimum records to keep a row ( MIN_N )             | 100                                |
| Maximum distinct TEST_NAME per group ( GROUP_SIZE ) | 100                                |
| Rows out                                            | 13,431 (9,211 distinct TEST_NAME ) |
| Groups formed                                       | 164                                |
| Group size — mean / median / min / max              | 56.2 / 57 / 1 / 100 names          |

**Sanity check.** Mean within-group pairwise string distance over 20 sampled groups: **10.18** for the distance-ordered groups vs **18.4** for groups formed by alphabetical order. Lower is more similar — the clustering is doing real work.

From here on, **every count is over codes with  $\geq 100$  records**. Everything below that floor is a long tail of one-off spellings that no amount of modelling fixes.

# Step 3 — Where our five codes landed

| TEST_NAME | UNIT     | group_id | group_path            |
|-----------|----------|----------|-----------------------|
| p-na      | mmol/l   | 42       | 2.1.1.2.1.2.1         |
| p-na      | (none)   | 42       | 2.1.1.2.1.2.1         |
| u-gluk-o  | (none)   | 79       | 2.1.2.2.2.1.1.2       |
| ab-ca-ion | mmol/l   | 89       | 2.1.2.2.2.2.1.1.2.2.1 |
| b-hba1c   | mmol/mol | 126      | 2.1.2.2.2.2.2.2.1.1   |

Both p-na rows sit in the same group, so the model sees the unit-bearing row and the unit-less row **in one prompt** and can reason about them together.

Source: DATA\_v4/3\_GroupCodesByStringDistance/knownInformationGrouped.tsv · the full dendrogram is in knownInformationGroupedStats.md

# Step 4 — Ask the model to name the LOINC

Each group goes to the LLM as one prompt. The model returns a **LOINC long common name** per row — not a code — so it is writing a description, not recalling an identifier it may hallucinate.

| TEST_NAME | UNIT     | group | loinc_name_guess                                       |
|-----------|----------|-------|--------------------------------------------------------|
| p-na      | mmol/l   | 42    | Sodium [Moles/volume] in Plasma                        |
| p-na      | (none)   | 42    | Sodium [Moles/volume] in Plasma                        |
| u-gluk-o  | (none)   | 79    | Glucose [Presence] in Urine by Test strip              |
| ab-ca-ion | mmol/l   | 89    | Calcium.ionized [Moles/volume] in Arterial blood       |
| b-hba1c   | mmol/mol | 126   | Hemoglobin A1c/Hemoglobin.total [Molar ratio] in Blood |

Both `p-na` rows get the **same** guess. This step names the *test*; it does not yet decide whether the evidence is sufficient. That decision is step 5's.

Example cached prompt: `DATA_v4/4_FindLOINC/groupsCache/42_prompt.md` · answer: `42.json` · 2,687 rows from 30 groups, USD 4.25

# Step 5 — Retrieve candidates, then ask again

The guessed name is run through **Hecate** (semantic search over the OMOP/LOINC vocabulary). The model is then shown the real candidates and must pick one — or decline.

Candidates for Sodium [Moles/volume] in Plasma:

| concept_id | concept_name                                    | score |
|------------|-------------------------------------------------|-------|
| 3019550    | Sodium [Moles/volume] in Serum or Plasma        | 0.934 |
| 3000285    | Sodium [Moles/volume] in Blood                  | 0.913 |
| 46235784   | Sodium [Moles/volume] in Serum, Plasma or Blood | 0.909 |
| 3041473    | Sodium [Moles/volume] in Venous blood           | 0.870 |

The model can only choose from concepts that **actually exist** in the vocabulary.  
Seven independent chances to be wrong (one per LOINC axis) are replaced by one choice from a real shortlist.

1,083 distinct guessed names needed candidates · 20,295 cached candidate rows · example prompt: [DATA\\_v4/5\\_FixLOINC/groupsCache/42\\_prompt.md](#)

## Step 5 — What it picked, and why

| TEST_NAME | UNIT     | picked concept                                   | certainty |
|-----------|----------|--------------------------------------------------|-----------|
| p-na      | mmol/l   | Sodium [Moles/volume] in Serum or Plasma         | high      |
| p-na      | (none)   | — declined —                                     | —         |
| u-gluk-o  | (none)   | Glucose [Presence] in Urine by Test strip        | high      |
| ab-ca-ion | mmol/l   | Calcium.ionized [Moles/volume] in Arterial blood | high      |
| b-hba1c   | mmol/mol | — declined —                                     | —         |

Two of the five are **declined on purpose**. An empty answer that says why is worth more than a guess that cannot be checked.

## Step 5 — Every answer carries its reasoning

Clause by clause, so the decision — or the refusal — can be read rather than guessed at.

**p-na [mmol/l]** → *"Sodium bcs LongName 'Natrium' ; [Moles/volume] bcs UNIT 'mmol/l' and value deciles [134..143] ; in Serum or Plasma bcs prefix 'p-' (Plasma), for which S/P is the correct LOINC system."*

**p-na []** → *"The property cannot be determined from the name alone. Without a unit or values, the choice between different properties is ambiguous."*

**u-gluk-o []** → *"[Presence] bcs LongName ends in '(kval)' and suffix is '-O' ; in Urine bcs prefix is 'U-' ; by Test strip bcs this is the standard method for qualitative urine glucose."*

**ab-ca-ion [mmol/l]** → *"Calcium.ionized bcs 'ab-ca-ion' ; [Moles/volume] bcs unit mmol/l and values ~1.0-1.2 ; in Arterial blood bcs prefix 'ab-'. "*

**b-hba1c [mmol/mol]** → *"Cannot map. The concept for HbA1c in IFCC units (mmol/mol), which has a [Substance Ratio] property, is not in the candidate list. "*

The last one is a **retrieval** failure, not a reasoning one — the model was right to decline. Reasoning text is what makes that distinction visible at all.

## Step 6 — Results: the funnel

| step                                | n_codes | p_codes | n_events   | p_events     |
|-------------------------------------|---------|---------|------------|--------------|
| total                               | 2,687   | 100.0%  | 51,790,415 | 100.0%       |
| has a guessed loinc                 | 2,577   | 95.9%   | 51,567,886 | 99.6%        |
| has a fixed loinc                   | 2,077   | 77.3%   | 47,581,148 | 91.9%        |
| exists in reference                 | 1,209   | 45.0%   | 50,528,414 | 97.6%        |
| agrees with reference (concept id)  | 742     | 27.6%   | 40,275,283 | <b>77.8%</b> |
| agrees with reference (LOINC Group) | 845     | 31.4%   | 41,457,043 | <b>80.0%</b> |

Read the **events** column. The codes the reference covers are the codes that carry the data: 45% of codes, but **97.6% of all records**. Agreement on **80% of records** is the number that matters operationally.

The reference ( `DATA_v4/ReferenceMappings/lab_data_summary.csv` , `APPROVED` rows) is the best mapping available, not ground truth — read these as *agreement*, not correctness.

# Step 6 — Results: by evidence level

| evidence_level   | n_codes | n_ai_mapped | n_agree | n_agree_group | p_ai_mapped | p_agree | p_agree_group |
|------------------|---------|-------------|---------|---------------|-------------|---------|---------------|
| name+unit+values | 590     | 570         | 428     | 485           | 96.6%       | 75.1%   | 85.1%         |
| name             | 441     | 292         | 177     | 210           | 66.2%       | 60.6%   | 71.9%         |
| name+values      | 138     | 136         | 107     | 114           | 98.6%       | 78.7%   | 83.8%         |
| name+unit        | 40      | 39          | 30      | 36            | 97.5%       | 76.9%   | 92.3%         |
| total            | 1,209   | 1,037       | 742     | 845           | 85.8%       | 71.6%   | 81.5%         |

**Message 1 — almost all the loss is name-only rows.** With a unit or values the pipeline answers 97-99% of the time. With a name alone it answers **66.2%** — it declines on purpose, because the unit is what fixes the *quantity*. `p-na [mmol/l]` maps; the same `p-na` with no unit does not.

# Step 6 — Results: what the disagreements are

| level                             | n_codes | % of checked | n_events   | % of events |
|-----------------------------------|---------|--------------|------------|-------------|
| agrees on the concept id (exact)  | 742     | 61.4%        | 40,275,283 | 79.7%       |
| agrees on the LOINC Group         | 845     | 69.9%        | 41,457,043 | 82.0%       |
| — of which recovered by the Group | 103     | 8.5%         | 1,181,760  | 2.3%        |

A **LOINC Group** is a value set of concepts that differ only in an axis the Group's rule rolls up — method above all. Two concepts in one Group are the same test measured differently. Membership is pulled from the OMOP vocabulary (`concept_class_id = 'LOINC Group'`), and the test is whether the two concepts **share** a Group.

**Message 2 — most disagreements are decoration, not the wrong analyte.** 103 of the 295 disagreements are the same test with a different specimen or method label. 90 of those share a Group with only 2-10 members, so the claim is tight.

# Step 6 — Results: disagreements that are really agreements

| TEST_NAME              | ours                                                | reference                                        | shared Group                                                                           | members |
|------------------------|-----------------------------------------------------|--------------------------------------------------|----------------------------------------------------------------------------------------|---------|
| ab-ca-ion              | Calcium.ionized ... in<br><b>Arterial blood</b>     | Calcium.ionized ... in<br><b>Serum or Plasma</b> | Calcium.ionized   Substance Concentration  <br>Moment in time   Blood, Serum or Plasma | 8       |
| u-ph-hy                | pH of Urine                                         | pH of Urine <b>by Test strip</b>                 | pH   Log Substance Concentration   Urine                                               | 8       |
| b-<br>neutrofiilit,... | Neutrophils ... <b>by</b><br><b>Automated count</b> | Neutrophils ...                                  | Neutrophils   Number Concentration (count/vol)  <br>Moment in time   Blood             | 3       |

ab-ca-ion is our running example: the pipeline reads the prefix ab- literally and says **Arterial blood**; the reference uses the generic **Serum or Plasma**. Same analyte, same property, same time — a specimen convention, not an error.

Reported with the Group's member count so a weak rollup can be discounted rather than taken on trust. Background: [RESEARCH/UnderstandingGroups.md](#)

## What is left

**Name-only rows.** 441 of 1,209 checked codes, and the pipeline declines on a third of them. The reference maps them anyway by convention; the pipeline holds out for evidence that will never arrive. This is a policy choice to revisit, not a modelling failure.

**Retrieval gaps.** `b-hba1c [mmol/mol]` has 1.98M records, a unit, and values — and still fails, because the IFCC `[Substance Ratio]` concept never appeared in the shortlist. The model was right to decline. The search was wrong to omit it.

**Where the Group score cannot help.** Only 737 of 1,209 checked codes carry a LOINC Group on both sides — about half the LOINC vocabulary has no Group at all (coagulation, cytology, CSF have no grouping rule). 80.0% is a **floor**, not a ceiling.

# Summary

**80.0% of records** agree with the curated reference

Almost all remaining loss is **rows with no unit**,  
and most remaining disagreement is **method/specimen decoration**

Reproduce: `STEPS/0..6/run.sh <DATA> --env build` · full report: `DATA_v4/6_EvaluateMapping/loincToOmopMappingStats.md` · background:  
`RESEARCH/UnderstandingGroups.md`